RESEARCH & DOCUMENTATION

StoryCast

Human stories. Shared with everyone.

A 3-page accessible microsite for audio and video stories about Rwanda, built with semantic HTML5, Sass and vanilla JavaScript.

Item	Detail
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Repository	https://github.com/g-teta/storycast.git
Figma	Wireframes, high-fidelity mockups and sitemap (links in section 9)

1. Project summary

StoryCast is a small website where people can listen to and watch short stories about Rwanda: a quiet moment on Lake Kivu, a basket weaver's morning, Kigali after dark. Its main idea is that every story is available in more than one way. Each one has a full written transcript, a clear label for audio or video, and a layout that works with a keyboard, a screen reader or a phone.

The site has three pages: **Home** (discover and filter stories), **Story Room** (one story with media, transcript and related stories) and **About** (mission and accessibility promise). It uses no frameworks. Styling is written in Sass and compiled to a single CSS file, and a small amount of vanilla JavaScript renders story data and handles the filter, the reading preview and the transcript download.

2. Research

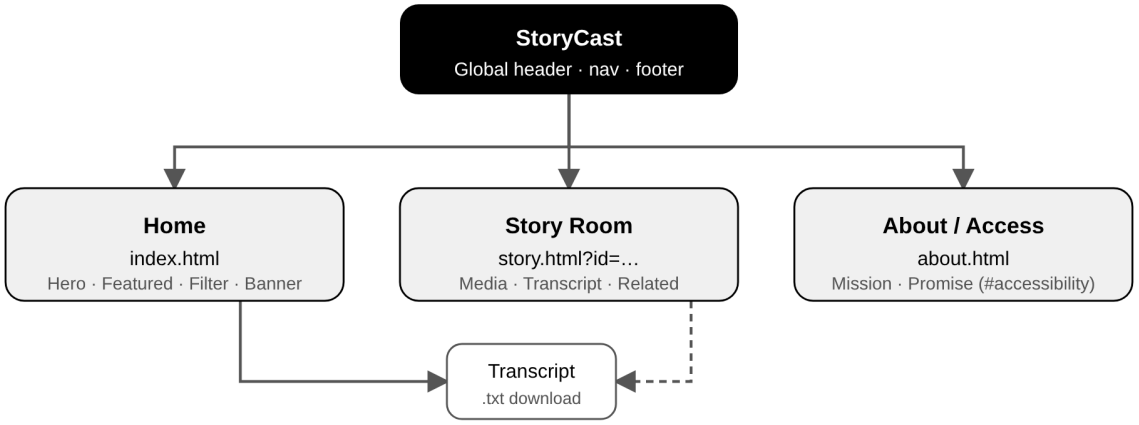
The research for this project was a review of the accessibility requirements that apply to audio and video content: WCAG 2.1 (W3C) and the W3C WAI guidance on time-based media. No interviews, surveys or user testing were carried out. The requirements below were used to decide what the site must include.

WCAG 2.1 criterion	How StoryCast applies it
1.2.1 Audio-only and video-only (A)	Every story has a full text transcript.
1.2.2 Captions, prerecorded (A)	The embedded video loads with captions turned on (cc_load_policy=1).
1.3.1 Info and relationships (A)	Structure uses real landmarks and headings.
1.4.3 Contrast, minimum (AA)	Text colours checked against 4.5:1 (see section 6).
2.1.1 Keyboard (A)	All controls are native links and buttons.
2.4.1 Bypass blocks (A)	A skip link on every page.
2.4.7 Focus visible (AA)	A visible focus ring on interactive elements.
4.1.3 Status messages (AA)	Filter results are announced through a live region.

The stories are set in Rwanda (Lake Kivu, basket weaving, Kigali in the evening). The three stories are sample content written for this project. The audio on the Story Room is a browser voice reading the transcript, not a recorded narration.

3. Information architecture

3.1 Sitemap



Solid arrows: primary navigation. Every page links to every other page through the header and footer.
Home cards ↔ Story Room (card links in, related stories link out to other stories).

Figure 1. StoryCast sitemap (editable version in Figma).

3.2 Page hierarchy

Page	Heading outline	Landmarks
Home	h1 Stories worth slowing down for · h2 Featured story title · h2 Find your next story · h3 each card · h2 Great stories are for everyone	header, nav, main (4 sections), footer
Story Room	h1 story title · h2 The full transcript · h2 Keep your curiosity going · h3 each related card	header, nav, breadcrumb nav, main, footer
About	h1 Human stories. Shared with everyone. · h2 A space for curiosity · h2 Every story, more ways to experience it · h3 three promise panels · h2 Your next perspective	header, nav, main, footer

3.3 Navigation and content model

The main navigation has three items (*Discover*, *About StoryCast*, *Accessibility*) and is identical on every page, with `aria-current="page"` marking the current location. Each story is one object with `id, type, category, duration, title, summary, author, alt, image, transcript[]` and optionally `video`. Cards on Home, the featured block and related stories on the Story Room are all generated from the same data, so a new story is added in one place.

4. Design

4.1 Concept

The design uses an off-white background, dark green and one terracotta accent colour, with large rounded photo frames. Wireframes were drawn first in Figma to set the structure, then developed into high-fidelity mockups of all three pages.

4.2 Design tokens

Token	Value	Use
$bg	#FAF7F2	Page background
$ink	#1F2A24	Body text, primary buttons
$muted	#5F6B64	Secondary text
$accent	#B4532A	Eyebrows, highlights
$dark	#1F3A2E	Banner, footer
$light / $line	#F4EFE6 / #E3DDD2	Chips, panels, borders
$font-serif / $font-sans	Playfair Display / Inter	Headings / body
$radius, $radius-media, $container-max, $gutter, $bp-tablet	24px, 28px, 1280px, 48px, 900px	Shape and layout

4.3 Responsive behaviour

Layouts use CSS Grid for page sections and card grids and Flexbox for the header, filters and button rows. At widths below 900px (the `$bp-tablet` breakpoint, applied through a `respond` mixin) multi-column layouts collapse to a single column and spacing tightens.

5. Development

5.1 Technology

HTML5, Sass (compiled with the `sass` npm package), and vanilla JavaScript. No frameworks or libraries. Fonts come from Google Fonts.

5.2 Project structure

```
storycast/
├── index.html           Home
├── story.html           Story Room (story.html?id=kivu-quiet)
├── about.html           About / Access
├── script.js            Story data + page logic
├── pictures/            Local photos
├── scss/
│   ├── main.scss        Entry point (@use of all partials)
│   ├── _variables.scss  Colour, type, shape, breakpoint tokens
│   ├── _mixins.scss     respond(), eyebrow, gradient
│   ├── _base.scss       Reset, typography, skip link, focus
│   ├── _layout.scss     Container, header, footer, grids
│   ├── _components.scss Buttons, chips, media frames, cards
│   └── pages/           _home, _about, _story
├── css/styles.css       Compiled output
└── package.json         build / watch scripts
```

5.3 Semantic HTML

Every page has a skip link, `header`, labelled `nav`, a single `main` and a `footer`. Content blocks are `section` elements labelled with `aria-labelledby` pointing to their heading; story cards and promise

panels are `article` elements; the story title block uses a `header`. Headings follow a strict h1 → h2 → h3 order.

5.4 JavaScript behaviour

- **Home:** renders the featured story and cards; the All / Audio / Video filter uses buttons with `aria-pressed` and writes the result count to a `role="status"` live region.
- **Story Room:** reads `?id=` from the URL, falls back to the first story, updates the page title, shows either the photo or a lazy-loaded video embed, prints the transcript and related stories.
- **Reading preview:** uses the browser's `speechSynthesis` to read the transcript aloud, with a progress bar exposed as `role="progressbar"` and a status note if the browser lacks support.
- **Transcript download:** builds a plain-text file in the browser with `Blob` and a temporary link.
- **Images:** each photo sits on a coloured gradient, so if a file fails to load the layout and description remain.

5.5 Running locally

```
npm install        # installs Sass
npm run build      # scss/main.scss → css/styles.css
npm run watch      # recompile on change
```

Then open `index.html` in a browser, or serve the folder with VS Code Live Server (port 5501 is preconfigured).

6. Accessibility checklist

Requirement	How it is met	Where
Skip navigation	First focusable element on every page, jumps to #main	All pages
Landmarks and headings	header, nav, main, section, article, footer; one h1; no skipped levels	All pages
Transcripts	Full transcript for all three stories on the page, plus .txt download	Story Room
Captions	YouTube embed loads with captions on (cc_load_policy=1); iframe has a descriptive title	Story Room (video)
Text alternatives	Photo frames use role="img" with a descriptive aria-label; the decorative img has empty alt	Cards, Story Room, About
Keyboard	Native links and buttons only; logical tab order; card image links removed from tab order to avoid duplicates	All pages
Focus visible	Custom focus ring on links, buttons and filters	_base.scss
ARIA where needed	aria-pressed on filters, aria-current on nav, role="status" live regions, role="progressbar" with value attributes, aria-label on nav and groups	Home, Story Room
Contrast	Text colours verified, see below	Palette
Motion	prefers-reduced-motion turns off transitions	_base.scss
Responsive and zoom	Fluid containers, single-column below 900px, relative text sizing	All pages
Language	lang="en" on every page	All pages

Contrast results (WCAG 2.1 relative luminance)

Foreground / background	Ratio	AA normal text (4.5)
Ink #1F2A24 on bg #FAF7F2	13.89:1	Pass
White on ink (primary button)	14.84:1	Pass
Banner text #D5E0D9 on #1F3A2E	9.10:1	Pass
Muted #5F6B64 on bg #FAF7F2	5.21:1	Pass
Muted #5F6B64 on light #F4EFE6	4.86:1	Pass
Accent #B4532A on bg #FAF7F2	4.67:1	Pass
Accent #B4532A on light #F4EFE6	4.35:1	Fails for small text, passes for large text (3:1)

7. Known limitations

- The audio on the Story Room is a browser voice reading the transcript, not recorded narration; real audio would use `<audio>` with a `<track>` and a downloadable transcript file.
- The accent colour on the light panel background (4.35:1) is under the AA requirement for small text.
- The layout adapts with one media query at 900px. The site does not yet use container queries, cascade layers or BEM class names.
- All stories are shown by one `story.html` page, which needs JavaScript to display a story.
- The site has not been tested with real users or assistive technology.

8. Tools

Figma (wireframes, mockups, sitemap), VS Code, Node.js and npm, Sass, Git and GitHub.

9. Links

- High-fidelity mockups: https://www.figma.com/design/XNtlrUcT9TNW0SdN6VbmPI/StoryCast?node-id=0-1&t=t5C9ZtPz7eM1E7sN-1
- Sitemap: https://www.figma.com/design/XNtlrUcT9TNW0SdN6VbmPI/StoryCast?node-id=4-2&t=t5C9ZtPz7eM1E7sN-1
- Wireframes: https://www.figma.com/design/XNtlrUcT9TNW0SdN6VbmPI/StoryCast?node-id=4-3&t=t5C9ZtPz7eM1E7sN-1
- Code: GitHub repository **https://github.com/g-teta/storycast.git**